

Two complete halves

Structure → process → surface consequence → qualification

STAGE 1

CRUST, ROCKS AND CRUSTAL DYNAMICS

STAGE 2

INDIA'S GEOLOGICAL ARCHITECTURE

EXAM LOGIC

- Build world foundations first.
- Then read India as shield–orogen–foreland.
- Practice, optional depth and final register complete the route.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Geological structure is the age, composition and architecture of rocks and tectonic features beneath a region, while physiography is their broad surface expression and geomorphology explains the processes that reshape it.

Crust versus lithosphere

Structure → process → surface consequence → qualification

continental crust

oceanic crust

rigid lithosphere

weaker asthenosphere

EXAM LOGIC

- Mineral and rock are different units.
- Crust is compositional; lithosphere is mechanical.
- Structure is modified by process at the surface.

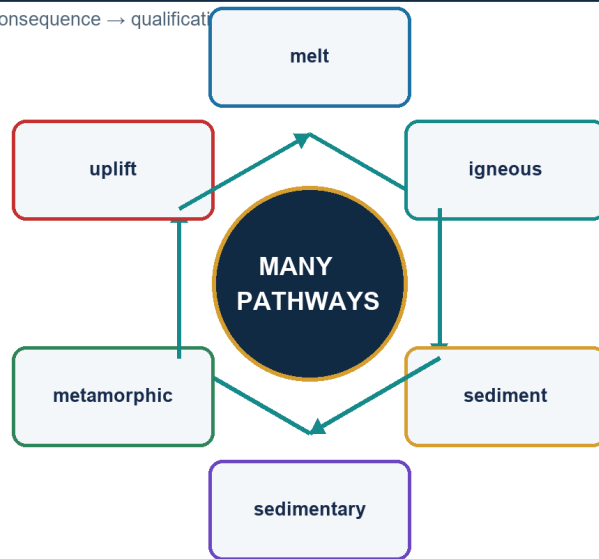
Geography 02 · original schematic · not to scale

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The crust is Earth's outer compositional shell, whereas the lithosphere is the stronger mechanical layer comprising the crust plus the rigid uppermost mantle.

Complete rock cycle

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Many pathways connect the three families.
- Uplift and exposure reopen the surface route.
- Metamorphism normally precedes melting.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The rock cycle is an open network in which melting, crystallisation, weathering, lithification, metamorphism and uplift can connect any rock family through multiple pathways rather than one fixed sequence.

Igneous rocks

Structure → process → surface consequence → qualification

KEY 1

intrusive ↔ extrusive

KEY 2

felsic ↔ intermediate ↔ mafic

KEY 3

texture records cooling

EXAM LOGIC

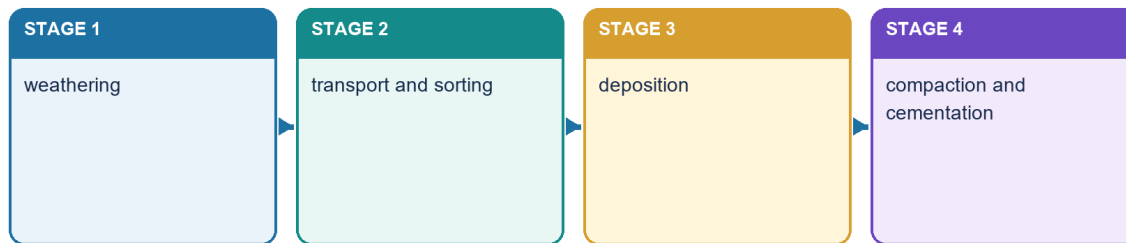
- Cooling rate controls crystal size.
- Granite–rhyolite, diorite–andesite and gabbro–basalt are broad pairs.
- Primary is genetic, not an age guarantee.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Igneous rocks record the cooling history and composition of magma or lava: slow intrusive cooling favours coarse crystals, while rapid extrusive cooling favours fine, glassy or vesicular textures.

Sedimentary rocks

Structure → process → surface consequence → qualification



EXAM LOGIC

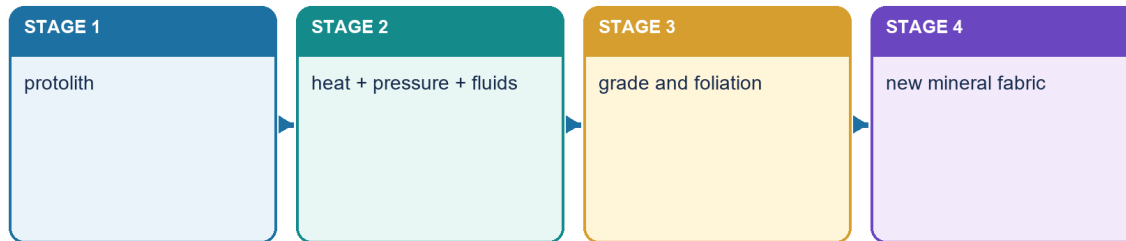
- Weather → transport → deposition → lithification.
- Clastic, chemical and biogenic routes differ.
- Field structures preserve environmental evidence.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Sedimentary rocks convert weathered, transported or chemically and biologically produced material into layered rock through deposition, compaction and cementation, while preserving clues such as fossils, ripples and mud cracks.

Metamorphic rocks

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Solid-state change by heat, pressure, stress and fluids.
- Foliation is not bedding.
- Grade paths are common, not universal.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Metamorphism changes a protolith in the solid state through heat, pressure, differential stress and reactive fluids, producing new minerals and fabrics without requiring complete melting.

Rock properties and applied geography

Structure → process → surface consequence → qualification

KEY 1

porosity ≠ permeability

KEY 2

fractures and weathering

KEY 3

soil · aquifer · dam · foundation

EXAM LOGIC

- Porosity and permeability differ.
- Secondary permeability matters in hard rock.
- Engineering conclusions are site-specific.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Rock name alone never determines land use: porosity, permeability, fractures, weathering, structure and site conditions jointly control groundwater, soils, foundations, dams and slope behaviour.

Continental drift evidence

Structure → process → surface consequence → qualification

KEY 1

shelf-edge fit

KEY 2

rock continuity

KEY 3

fossils

KEY 4

palaeoclimate

KEY 5

palaeomagnetism

Geography 02 · original schematic · not to scale

EXAM LOGIC

- Use five independent evidence classes.
- Shelf edges outperform shorelines.
- Wegener's mechanism remained inadequate.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Continental drift became persuasive because shelf-edge fit, matching rock belts, fossils, palaeoclimate and palaeomagnetism independently point to former continental connections, even though Wegener lacked an adequate mechanism.

Sea-floor spreading

Structure → process → surface consequence → qualification

KEY 1

ridge magma and new crust

KEY 2

symmetrical magnetic stripes

KEY 3

older and deeper away from ridge

KEY 4

trench subduction

Geography 02 · original schematic · not to scale

EXAM LOGIC

- Ridge creation records magnetic polarity.
- Age and sediment generally increase away from ridges.
- Subduction closes the mass balance.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Sea-floor spreading creates young oceanic crust at mid-ocean ridges, records symmetrical magnetic stripes and balances creation by consuming older lithosphere at trenches, so the Earth does not expand.

Plate-boundary matrix

Structure → process → surface consequence → qualification

KEY 1

divergent

KEY 2

ocean–ocean / ocean–continent

KEY 3

continent–continent

KEY 4

transform

Geography 02 · original schematic · not to scale

EXAM LOGIC

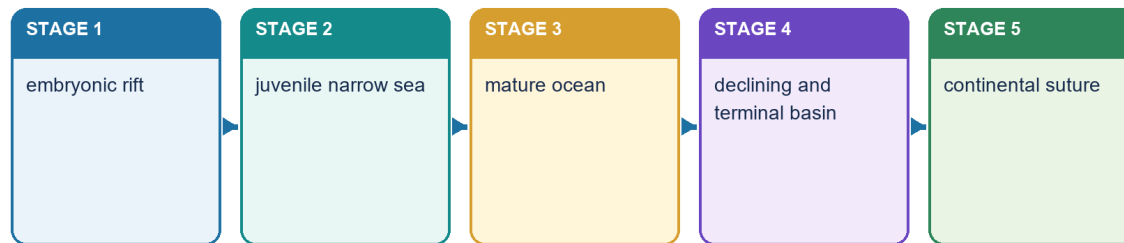
- Divergent, three convergent settings and transform.
- Andes and Himalaya require different mechanisms.
- Plate interiors can still be active.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Plate boundaries are diagnosed by motion and products: divergence creates crust, subduction destroys oceanic crust, continent collision thickens crust, and transform motion conserves crust while generating shallow earthquakes.

Wilson-cycle analogy

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Rift, narrow sea, ocean, closure and suture.
- Examples are present-day analogies.
- No basin is destined to finish the sequence.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The Wilson cycle is an analogy for rifting, ocean opening, subduction, closure and suturing, not a fixed destiny that every ocean basin must complete.

Isostasy and orogeny

Structure → process → surface consequence → qualification

KEY 1

Airy: crustal roots

KEY 2

Pratt: density contrast

KEY 3

loading and rebound

KEY 4

shortening and mountain building

Geography 02 · original schematic · not to scale

EXAM LOGIC

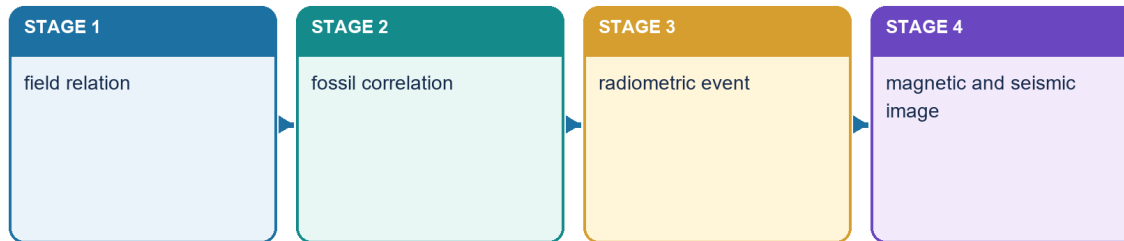
- Airy roots and Pratt density are exam-safe concepts.
- Loading and unloading drive adjustment.
- Mountain building and erosional response interact.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Isostasy explains vertical crustal adjustment, while orogeny explains mountain building through shortening, thickening and uplift; erosion and unloading can then trigger further isostatic response.

Geological time and evidence

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

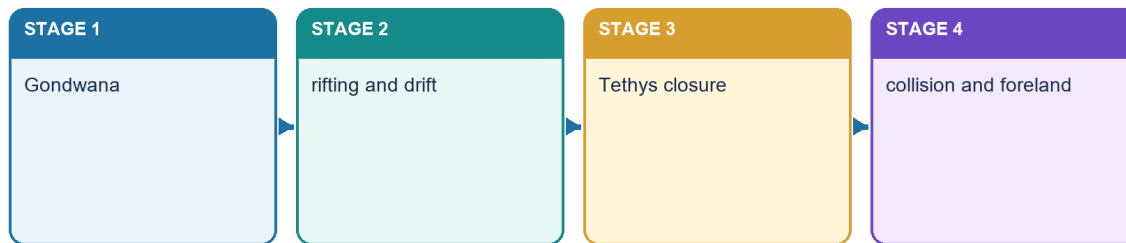
- Field relations come first.
- Dates record events, not automatic system ages.
- Geophysics images concealed structure.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

A safe geological chronology combines superposition, cross-cutting relations, unconformities, fossils, radiometric event ages, palaeomagnetism and geophysics; relative order is often safer than memorised numerical ages.

Indian plate tectonic journey

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Gondwana rift and northward drift.
- Tethyan closure and prolonged collision.
- Uplift, foreland flexure and seismicity continue.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The Indian plate journey links Gondwana rifting, northward motion, Tethyan ocean closure and prolonged India-Eurasia collision to Himalayan uplift, foreland subsidence and continuing seismicity.

India's three-province skeleton

Structure → process → surface consequence → qualification

Peninsular shield

foreland alluvium

Himalayan thrust belt

EXAM LOGIC

- Ancient Peninsular shield.
- Active Himalayan fold-thrust belt.
- Sediment-filled foreland plain.

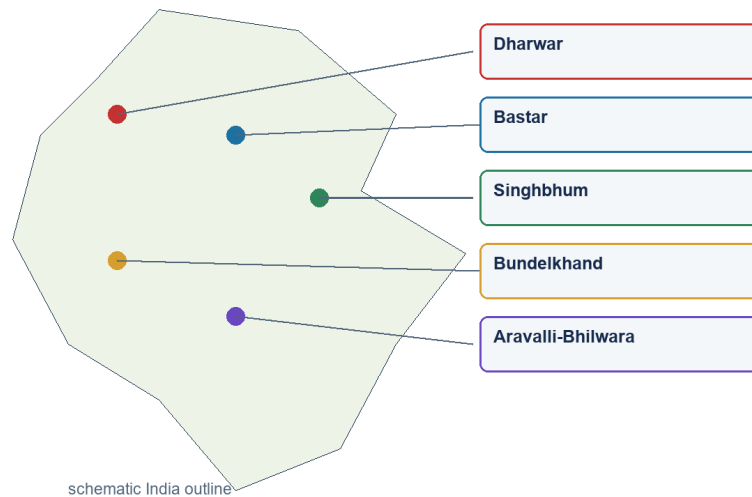
Geography 02 · original schematic · not to scale

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

India's geological skeleton consists of an ancient Peninsular shield, an active Himalayan fold-thrust belt and a sediment-filled Indo-Ganga-Brahmaputra foreland basin linked by collision, flexure, erosion and deposition.

Cratons and mobile belts

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

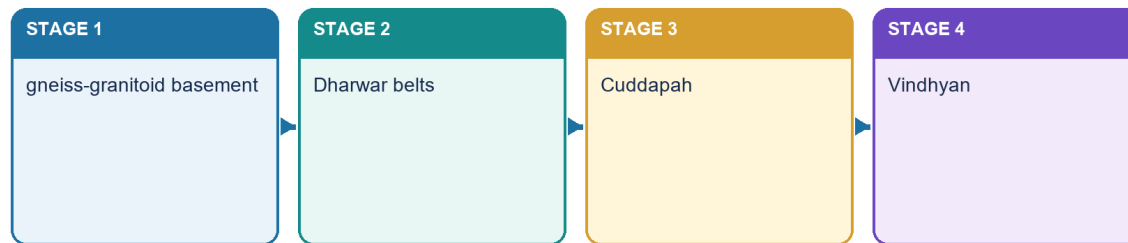
- Dharwar, Bastar, Singhbhum, Bundelkhand, Aravalli-Bhilwara.
- Later rifts, dykes, basins and lava complicate the shield.
- Stable never means uniform.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Peninsular India is a mosaic of cratons and mobile belts later cut by faults, rifts, dykes and basins and partly covered by younger sediments and lava, so stable never means uniform, flat or earthquake-free.

Ancient shield and Purana basins

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Basement differs from supracrustal belts.
- Cuddapah and Vindhyan overlie older crust.
- Ore and stone links require qualification.

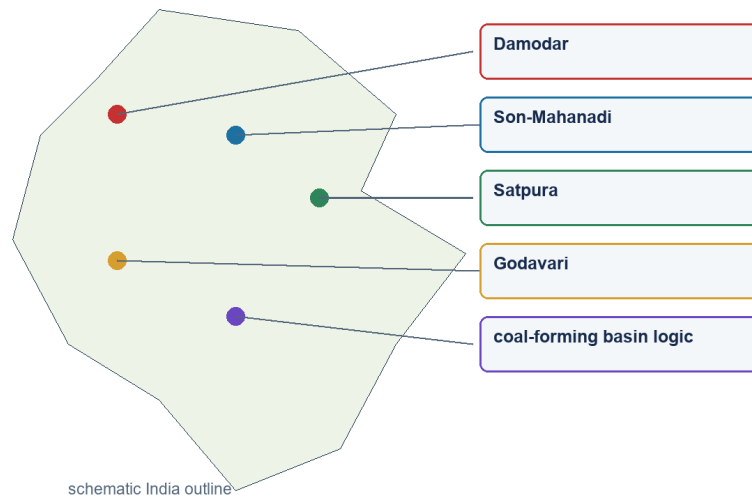
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Archaean granitoid-gneiss basement and Dharwar-type supracrustal belts form distinct parts of the ancient shield, and their ore associations reflect specific magmatic, sedimentary, metamorphic and hydrothermal histories rather than age alone.

Purana is an older Indian label for Proterozoic basin successions such as Cuddapah and Vindhyan, which rest unconformably on older basement and record long-lived sedimentation rather than one uniform mineral province.

Gondwana rift basins

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Fault-bounded continental basins.
- Damodar, Son-Mahanadi, Satpura and Godavari hooks.
- Coal needs environment, burial and preservation.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The Gondwana Supergroup is a fault-basin continental sedimentary succession with important coal measures; it must not be confused with Gondwana, the southern palaeocontinent.

Marine and offshore sedimentary basins

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Basin name alone is insufficient.
- Source, maturation, migration, reservoir, seal, trap and timing.
- Passive-margin architecture creates opportunity.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

A sedimentary basin offers petroleum potential only when source, maturation, migration, reservoir, seal, trap and timing form a complete preserved system, whether onshore or offshore.

Deccan Trap

Structure → process → surface consequence → qualification

stacked basalt flows

dykes

vesicular contacts

regur parent material

EXAM LOGIC

- Stacked Late Cretaceous basalt flows.
- Contacts, vesicles, fractures, dykes and intertrappeans.
- Relief, soil and aquifers vary by architecture.

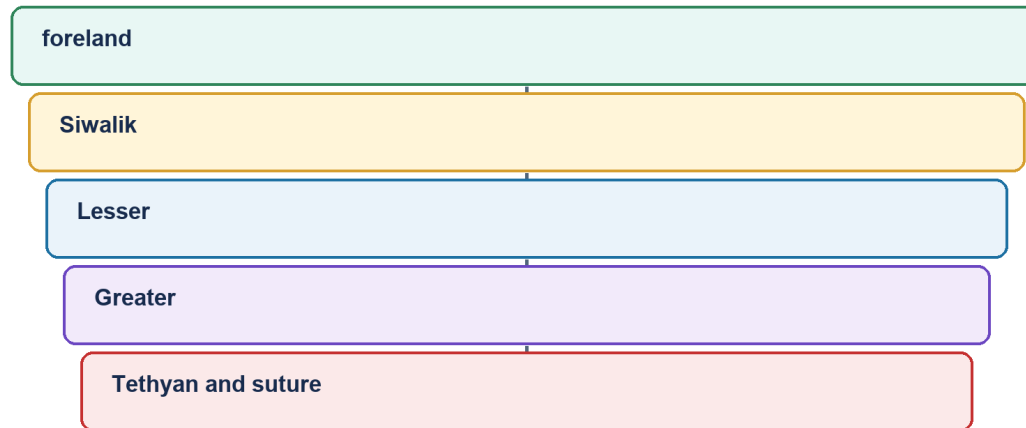
Geography 02 · original schematic · not to scale

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The Deccan Trap is a Late Cretaceous flood-basalt province of stacked fissure-fed flows whose contacts, fractures, vesicular zones and weathering create step relief, black-soil parent material and highly variable aquifers.

Himalayan structural order

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Foreland–Siwalik–Lesser–Greater–Tethyan/suture.
- Collision orogen, not Andes-type arc.
- Cross-section is conceptual and laterally variable.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The Himalaya is a continent-collision fold-thrust orogen containing sedimentary, metamorphic and crystalline belts; unlike the Andes, it is not primarily a volcanic subduction arc.

Foreland basin and alluvium

Structure → process → surface consequence → qualification

mountain load

plate flexure

coarse-to-fine fill

aquifer and flood links

EXAM LOGIC

- Mountain load bends the plate.
- Sediment fills accommodation.
- Aquifer, flood, subsidence and site response follow.

Geography 02 · original schematic · not to scale

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The Indo-Ganga-Brahmaputra Plain is not structureless alluvium but a flexural foreland basin whose thick sediment supports fertile soils and aquifers while also influencing floods, subsidence and seismic site response.

Coasts, offshore and islands

Structure → process → surface consequence → qualification

KEY 1

western rifted margin

KEY 2

eastern deltaic basins

KEY 3

Andaman arc

KEY 4

Lakshadweep atolls

Geography 02 · original schematic · not to scale

EXAM LOGIC

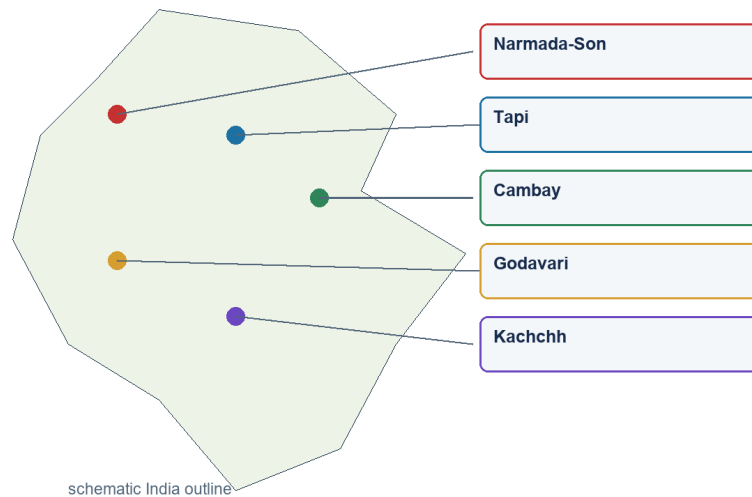
- Western rifted sectors and eastern deltaic basins.
- Andaman arc differs from Lakshadweep atolls.
- Island geology and hazards are not uniform.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Andaman-Nicobar belongs to an active subduction-related arc-forearc system, whereas Lakshadweep consists of coral atolls on a submarine volcanic-ridge foundation; Indian island geology is therefore not uniform.

Faults, rifts and lineaments

Structure → process → surface consequence → qualification



Geography 02 · original schematic · not to scale

EXAM LOGIC

- Narmada-Son, Tapi, Cambay, Godavari and Kachchh.
- Rift, graben and lineament are distinct.
- Mapped linearity does not prove active faulting.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

A rift is an extensional crustal zone, a graben is a down-dropped fault block, and a lineament is a mapped linear feature that is not automatically an active fault.

Structure to surface and resources

Structure → process → surface consequence → qualification

KEY 1

relief

KEY 2

drainage

KEY 3

groundwater

KEY 4

resources

KEY 5

bounded hazards

EXAM LOGIC

- Link structure to process before consequence.
- Geological association is not resource determinism.
- Seismic application remains bounded to Topic 03.

Geography 02 · original schematic · not to scale

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

Geological structure guides scarps, valleys, drainage patterns, waterfalls and groundwater through rock contrasts, bedding, joints and faults, but climate, erosion, base level and human action prevent one-to-one control.

Shield belts, Gondwana basins and sedimentary margins are associated respectively with many metallic ores, coal and petroleum, but exploration, grade, preservation, technology, economics and environmental limits decide usable resources.

India's seismicity reflects active Himalayan and Andaman plate boundaries plus reactivation of inherited structures such as Kachchh and selected Peninsular faults; seismic zoning is a design generalisation, not earthquake prediction.

Regional comparison, NCMM and answer method

Structure → process → surface consequence → qualification

KEY 1

Peninsula ↔ Himalaya ↔ plain

KEY 2

NCMM value-chain anchor

KEY 3

evidence → mechanism → consequence → limit

EXAM LOGIC

- Compare Peninsula, Himalaya and plain.
- PYQ routes: 2022 rocks; 2025 tectonic change; 2024/2025/2026 Prelims; adjacent Deccan/Gondwana.
- Prelims traps: Gondwana terms; basin != oil; stable != aseismic; mission != deposit.
- Answer spine: evidence -> mechanism -> consequence -> limitation -> verdict.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM

The National Critical Mineral Mission, approved on 29 January 2025, links domestic and offshore exploration with mining, processing, recycling, research, stockpiling and overseas acquisition, but mission targets do not prove any particular deposit or reserve.

India's surface geography is best explained as inherited geological structure filtered through tectonics, weathering, climate, erosion, sediment transfer and human use, so every high-scoring answer must connect evidence, mechanism, consequence and limitation.